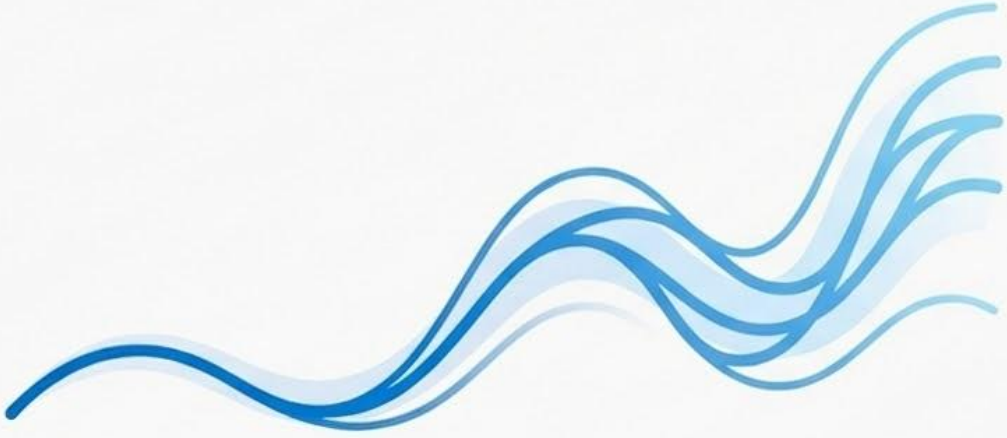

Zoom Momentum

Gamifying how students learn in online classes

NotebookLM

Virtual education suffers from two massive energy leaks

The “Cold Start”

The first 5–10 minutes of every Zoom class are “dead air.” Students sit in the Waiting Room or the main meeting in awkward silence, checking social media. By the time the professor starts, brains are disengaged.

The “Passive Learner” Effect

Attention drops significantly after 15 minutes of one-way audio. Without physical cues, students drift. They are often too embarrassed to ask for help when they get lost, compounding the disconnect.

NotebookLM

Zoom Momentum: A Dual-Ended Engagement Ecosystem

Before Class

Synchronous Activation

Transforms the Waiting Room from a holding pen into a **Synchronous Live Trivia Arena**, priming student brains before the camera even turns on.

During Class

Bi-Directional Intervention

Offers a **Private Recovery Mechanism** for students to self-correct, while giving professors a real-time “Pulse” on the room with one-click interventions.

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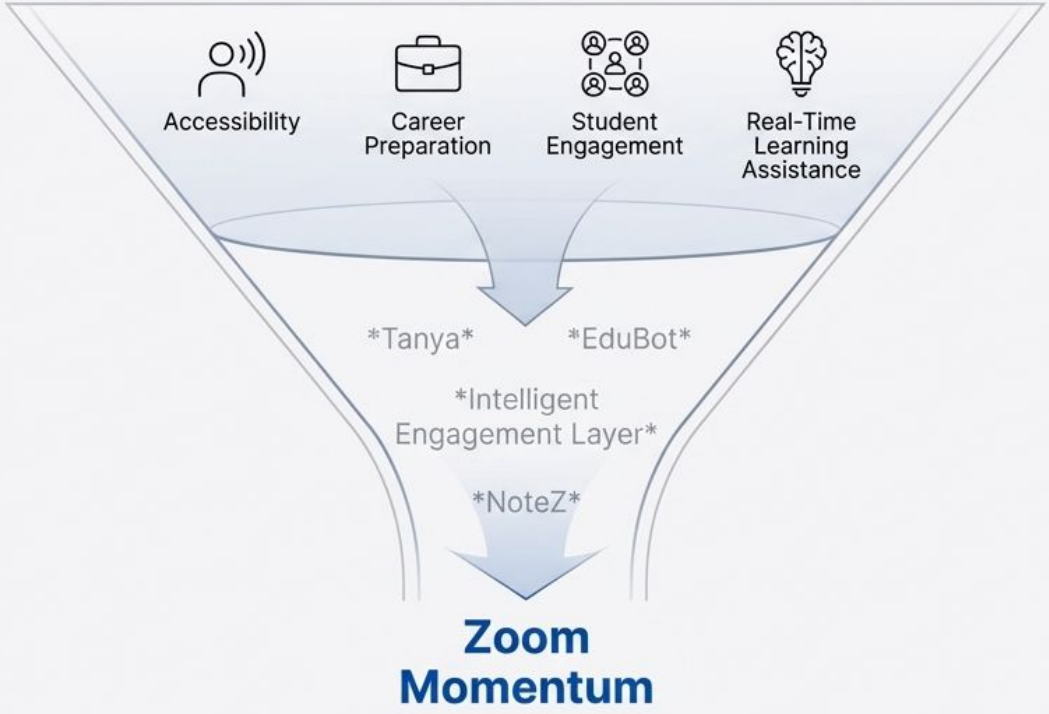
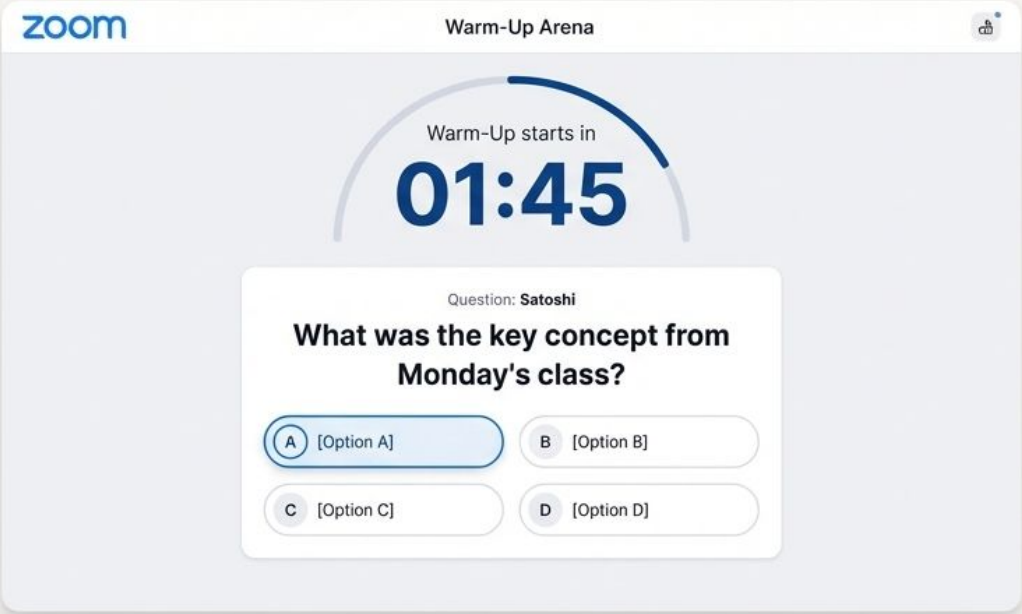
Momentum is Built on a Foundation of Community-Driven Innovation



Momentum synthesises the most promising, feasible, and high-impact concepts from our hackathon research into a single, cohesive ecosystem. It addresses the most persistent needs expressed by students and educators.

NotebookLM

Feature 1: The Warm-Up — Turning Wait Time into Engaged Time



The User Experience

9:55 AM

Student joins and enters the live trivia lobby instead of a static "Please Wait" screen.

9:58 AM

The game triggers for everyone simultaneously, mixing fun icebreakers with academic recall questions.

10:00 AM

As the professor begins, the "Top 3 Warm-Up Leaders" are flashed on screen, injecting **immediate** high energy into the class.

Hackathon DNA

Inspired by: **Tanya* (Gamified points & leaderboards) + **Student Personalization** (Recalling past context).

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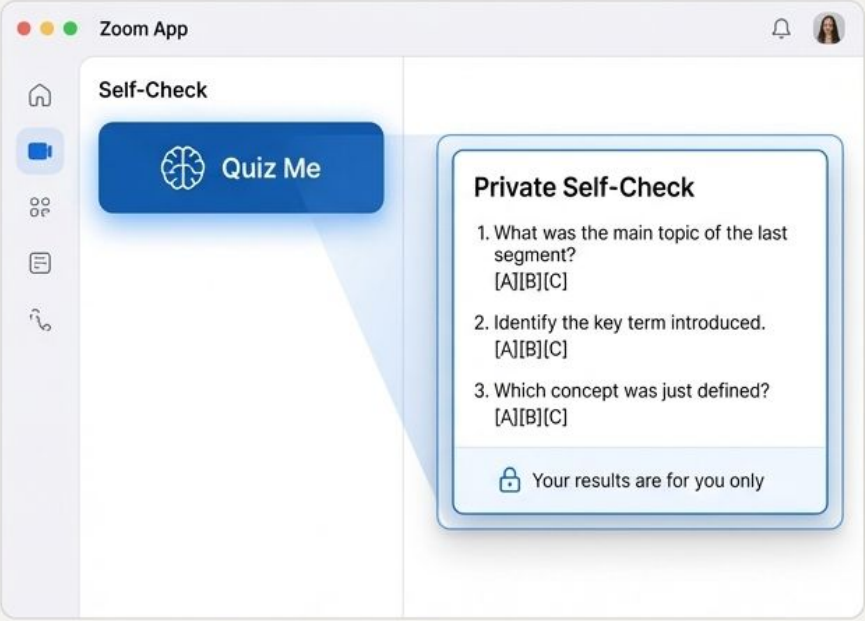
Feature 2: The Self-Check — A Private Path to Recovery



The User Experience

The Pain Point
A student zones out and realizes they're lost, but is too embarrassed to interrupt and ask a question.

The Action
They click the private 'Quiz Me' button.

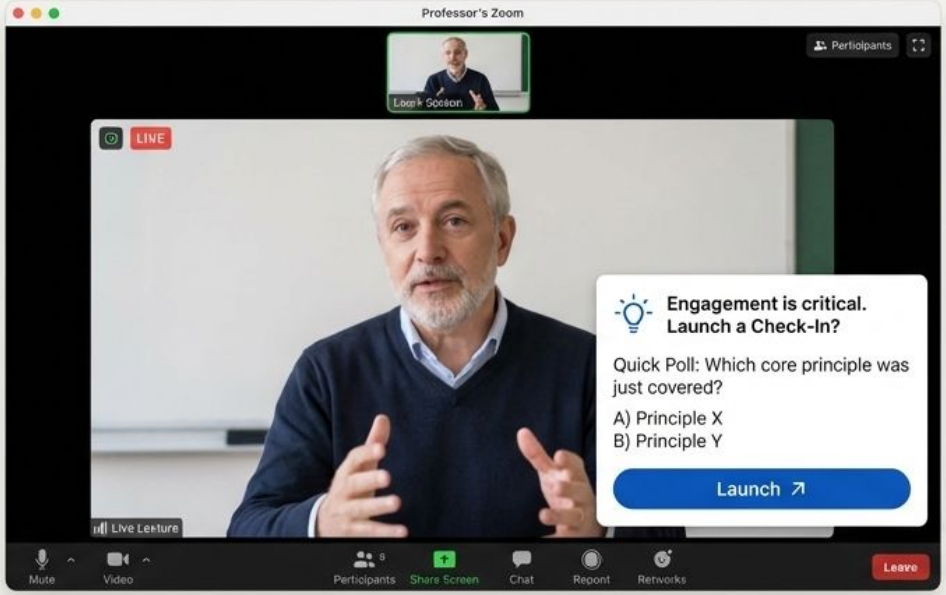
The Result
The AI generates a quick quiz. The student's results are **Local Only** and are not sent to the professor. This 'Safe Failure' allows them to realize they need to refocus without fear of penalty or embarrassment.

Hackathon DNA

Inspired by: ***EduBot*** (Adaptive, on-the-spot knowledge checks) + ***NoteZ*** (Real-time transcript analysis).

NotebookLM

Feature 3: The Professor's Pulse — An AI Co-Pilot for Engagement



Hackathon DNA

Inspired by: *Intelligent Engagement Layer* (Metadata signal monitoring) + *Tanya* (Automated polling).

The User Experience

Sensing

The system monitors non-invasive, passive signals (e.g., chat velocity drops to zero, a screen share remains static for 10 minutes).

Nudging

The professor receives the private toast notification with a suggestion.

One-Click Fix

The professor clicks 'Launch," instantly deploying an AI-generated poll based on the current topic, all without breaking their teaching flow.

NotebookLM

Momentum is More Than a Feature; It’s a Strategic Evolution for Zoom

1. Activate Unused Digital Real Estate

The Waiting Room is currently a wasted asset. The Warm-Up transforms it into a valuable, **high-engagement space** that strengthens the user experience from the very first touchpoint.

2. Champion a Privacy-First AI Narrative

While competitors explore invasive eye-tracking, Momentum’s use of aggregated system signals is a powerful differentiator. This positions Zoom as a leader in responsible, FERPA-compliant AI innovation.

3. Evolve AI Companion from Secretary to Teaching Assistant

Currently, AI Companion is a brilliant secretary that takes notes and summarizes. Momentum elevates it to a **proactive teaching assistant** that proctors quizzes, senses room energy, and supports pedagogy.

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Driving Tangible Educational Outcomes

For Students

- **Enables "Safe Failure":** The private "Self-Check" allows students to identify and correct knowledge gaps during class, when they can still recover, rather than after a midterm.
- **Reduces Cognitive Disconnect:** The "Warm-Up" primes the brain with relevant content, ensuring students enter the lecture ready to learn.
- **Builds Confidence:** By providing a low-stakes way to participate and self-assess, Momentum helps reduce the anxiety anxiety common in virtual learning environments.

For Universities

- **Boosts Retention:** Students who engage in the first 5 minutes are **3x more likely to stay focused**. Active engagement is a leading indicator of student success and retention.
- **Provides Actionable Pedagogical Insight:** Aggregated, anonymous data from the "Professor's Pulse" helps instructors **understand which parts of their lectures are least engaging** and **refine their methods**.

NotebookLM

A Phased Approach to Building Momentum

1

Phase 1: Prototype

Goal: Build a functional proof-of-concept focusing on The Warm-Up and Self-Check features using the Zoom Apps SDK and AI Companion APIs.

2

Phase 2: Internal Pilot with ASU

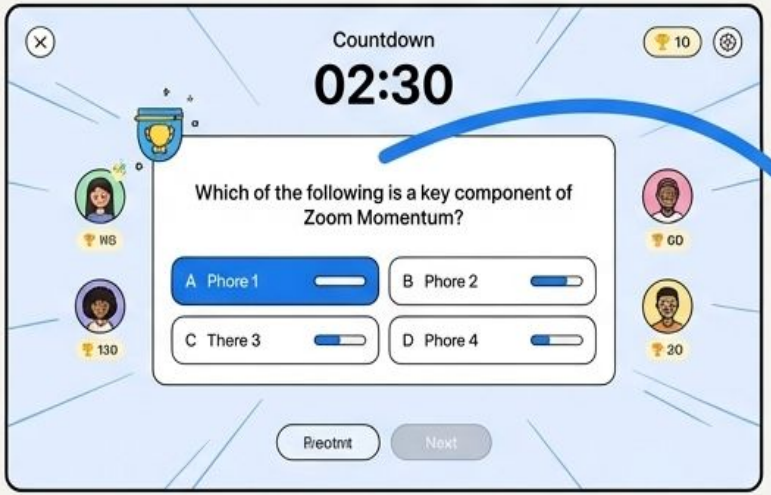
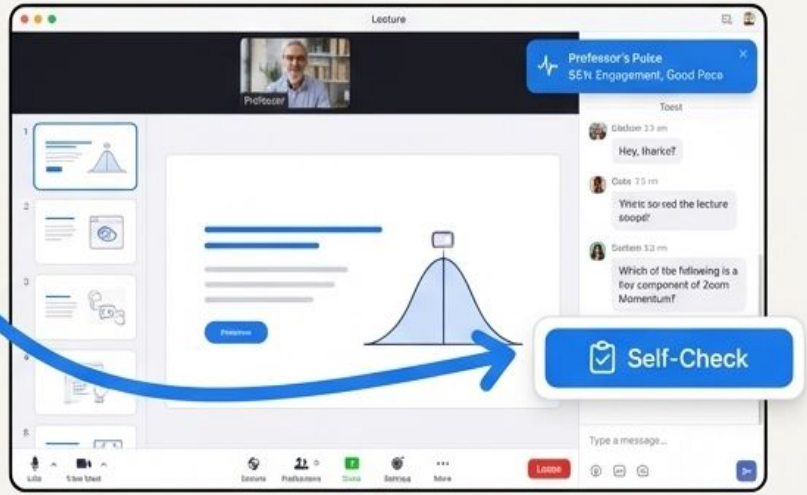
Goal: Deploy with a select group of educators to gather real-world data and refine the "Professor's Pulse" signal thresholds and UI/UX based on direct feedback.

3

Phase 3: Platform Integration

From the First Minute to the Last, Momentum Keeps the Connection Alive

Warm-Up Arena



Lecture View